Free-body diagrams

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Draw only the external forces on one chosen object.

SUCCESS CRITERIA - I can...

1. Choose one object and list its external interactions.
2. Draw force arrows in physically correct directions.
3. Remove velocity and net force as extra interaction arrows.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A fixed-mass cart has acceleration 3 m/s² under 6 N net force. Find acceleration when net force becomes 10 N. Explain.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Sigma F_x = ma_x$; $\Sigma F_y = ma_y$; $F_g = mg$.
Draw only forces on the chosen object. x and y name directions. Use g = 9.8 N/kg unless another value is given.

Worked reasoning: A lamp hangs motionless from one vertical cable. With weight 12 N down, the cable tension is 12 N up. Only forces acting on the lamp belong in its diagram.

Your turn: Draw the lamp as a labeled point with its two force arrows. Explain why its force on the cable is not another arrow on this diagram.

Add and label force arrows.

Lamp: draw the two 12 N forces acting on it.

Marzano check - circle: ① ② ③

03 A book at rest

Draw and label the forces on a book resting on a level table. Name both sources. Use arrows whose comparative lengths match rest.

Add and label force arrows.

The dot represents the object. Object: book

Marzano check - circle: ① ② ③

04 After the hand lets go

A box slides to the right on a rough floor after release. Draw its forces. Which way does friction act?

Add and label force arrows.

Box slides right after the hand lets go.

Marzano check - circle: ① ② ③

05 A falling ball

A ball falls down and is speeding up. Air pushes up. Draw weight and air resistance. Which arrow should be longer?



Add and label force arrows.

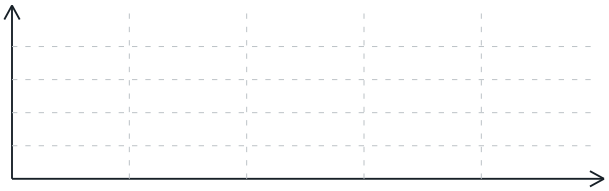
Falling ball speeds up; air resistance acts upward.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A cart goes from 18 m to 6 m in 4 s at steady velocity. Find velocity and sketch its position-time graph. Label the model with the given positions and any prediction.

Position (m)



Time (s)

Marzano check - circle: 1 2 3

07 Exit check

A ball moves upward after leaving a hand. Ignore air resistance. Draw its free-body diagram and justify every arrow.



Add and label force arrows.

Ball travels upward after leaving the hand.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Draw separate diagrams for a hanging lamp at rest and the same lamp moving upward at steady speed. What can a force diagram alone fail to tell you?



Lamp at rest



Lamp moving up steadily

Marzano check - circle: 1 2 3